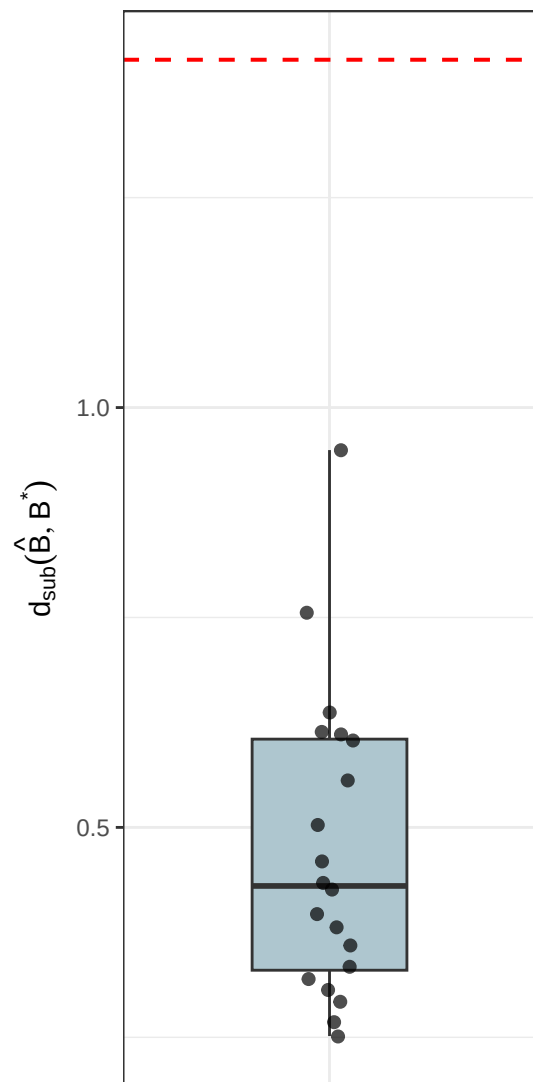


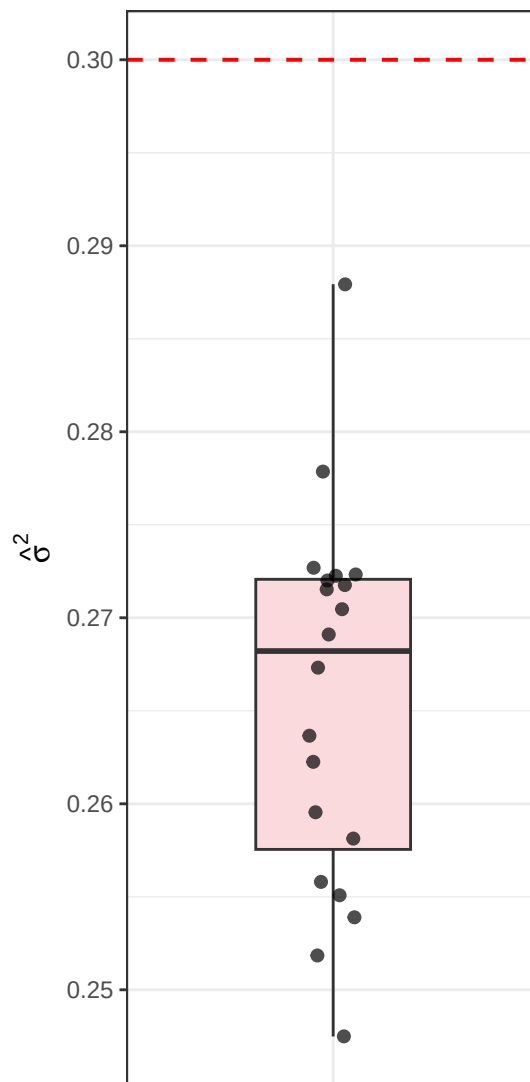
# Algorithm robustness across 20 independent replications

DGP fixed:  $N=800$ ,  $Q=50$ ,  $J=30$ ,  $K=2$ ,  $\sigma^2_{\text{true}}=0.3$

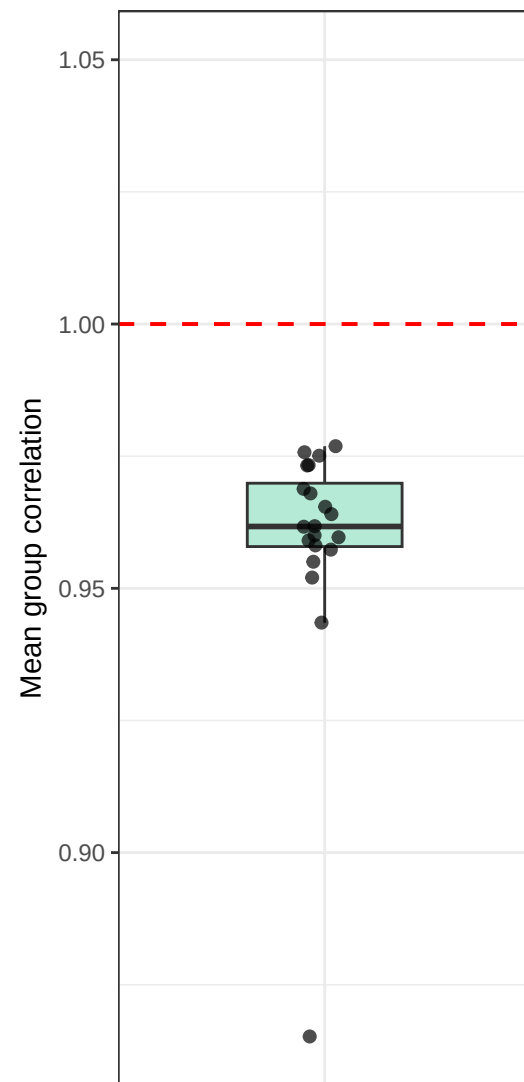
B subspace distance



$\hat{\sigma}^2$  estimate



$\bar{r}(\hat{\alpha}, \alpha^*)$



Dashed: worst-case bound  $\sqrt{K} = 1.414$

Dashed: true value 0.3

Dashed: perfect recovery